STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 12: Kinetic Theory

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Equivalent correct methods should receive credit.

Q1 1 mark

Negligible molecular volume and forces except during elastic collisions.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q2 1 mark

pV=nRT.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q3 1 mark

3k_BT/2.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q4 1 mark

Molecular momentum transfer to walls.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q5 1 mark

Three translational.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q6 1 mark

It increases.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q7 1 mark

3R/2.

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q8 1 mark

sqrt(3RT/M).

1 mark for the correct fact, unit or formula. Accept an equivalent correct method.

Q9 2 marks

Doubles.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q10 2 marks

About 300 K.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q11 2 marks

~6.21 x 10-^2 J.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q12 2 marks

2.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q13 2 marks

3nRT/2.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q14 2 marks

Negligible particle volume and negligible intermolecular forces except collisions.

Suggested split: 1 mark for the principle/formula and 1 for the answer or explanation. Accept an equivalent correct method.

Q15 3 marks

~517 m/s.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q16 3 marks

~249 J.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q17 3 marks

One-third.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q18 3 marks

Three translational modes contribute 3k_BT/2 per molecule.

Suggested split: 1 for the relation, 1 for working/reasoning and 1 for result with unit. Accept an equivalent correct method.

Q19 4 marks

Lighter gas speed is twice as high.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.

Q20 4 marks

U=(5/2)nRT~12.5 kJ.

Suggested split: 1 for principle, 1 for equation, 1 for working and 1 for result/explanation. Accept an equivalent correct method.